

Python for Web development Syllabus

Module 1: Python Fundamentals

1. Python Basics

- What is Python?
- Installing Python & Jupyter
- First Python Program

2. Core Python Constructs & Functions

- Variables and Data Types
- Lists, Tuples, Sets, Dictionaries, If, else, elif conditions, Loops (for, while)
- Defining and calling functions, Parameters and Return values
- Lambda functions

3. File Handling, Exception Handling & Object-Oriented Programming

- Reading and writing files
- Try-Except-Else-Finally blocks
- Classes & Objects
- Inheritance & Encapsulation

4. Modules, Packages & Regular Expressions

- Built-in modules: os, sys, math
- Creating your own modules
- Pattern matching and searching
- re.match(), re.search(), re.sub()

Module 2: Frontend Basics (Just Enough to Build Web UIs)

1. HTML Essentials

- 50+ Tags, 40+ Attributes
- Forms, Inputs, Buttons
- Tables, Divs, Spans

2. CSS & Bootstrap (Overview)

- Basic styling
- Using Bootstrap components

Module 3: Web Development with Flask

1. Flask Introduction

- Flask vs Django
- Installation & Setup
- App structure and execution

2. Routing, HTTP Methods, Templates & Jinja2

- URL routes, GET vs POST, Route variables & render_template()
- Template inheritance & Passing data with Jinja2

3. Forms, User Input, Session & Cookies

- Handling form data, Validation & File uploads
- session[] & secret_key
- Auth basics (login/logout)

4. Database Integration

- SQLite/MySQL with mysql.connector or SQLAlchemy ORM
- CRUD operations

5. Static Files

- Serving CSS, JS, Images & Uploading user files

6. Advanced Flask & Flask extensions (overview)

- Blueprints, Flash messages & Pagination
- Custom error pages, Flask Mail & Flask WTF
- Flask SQLAlchemy : Simplified ORM integration for Flask apps

7. Deployment

- WSGI setup
- Hosting on PythonAnywhere or Heroku
- Using environment variables

Module 4: Django Basics (& Advance)

1. Django Setup & Models

- Project vs App, Admin panel
- URL routing & Templates
- ORM basics, Migration
- CRUD operations

2. Django Forms & Authentication

- Model forms & Validation
- Registration/Login, User roles